

Statistical Properties of the Sharpe Ratio Estimator

A Robustness Analysis under Various Assumptions on Returns

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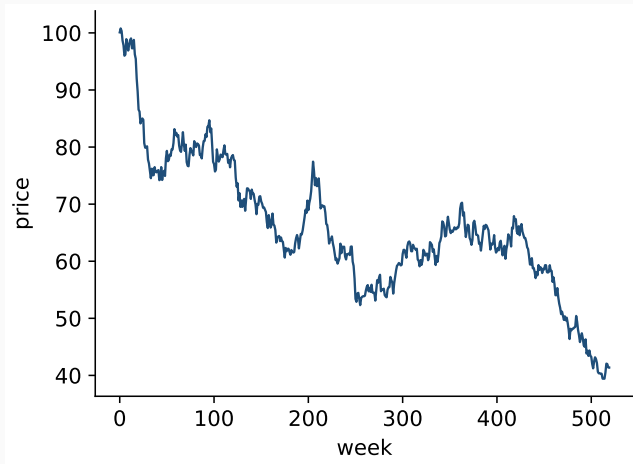
**How much can you trust a
Sharpe ratio?**

How reliable is this number?

- A reported Sharpe ratio of $\widehat{SR} = 0.5$.
- A population quantity, estimated from a finite sample:

$$SR = \frac{\mu}{\sigma}, \quad \widehat{SR} = \frac{\hat{\mu}}{\hat{\sigma}}$$

- One question for the whole talk: **how much should we trust it?**

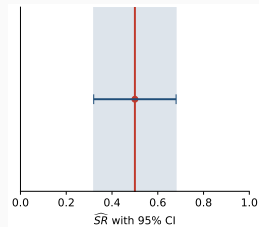
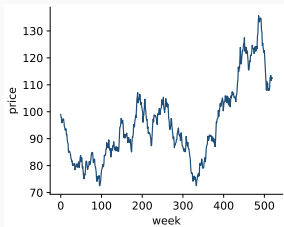


Reported $\widehat{SR} = 0.5$.

Measuring reliability: the asymptotic variance

$$\sqrt{T} (\widehat{\text{SR}} - \text{SR}) \xrightarrow{d} \mathcal{N}(0, V_{\widehat{\text{SR}}})$$

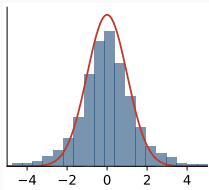
Classical IID–Normal result (Lo, 2002): $V_{\widehat{\text{SR}}}^{\text{IID N}} = 1 + \frac{1}{2} \text{SR}^2$ CI: $\text{IC}_{1-\alpha} = \left[\widehat{\text{SR}} \pm z_{1-\alpha/2} \sqrt{V_{\widehat{\text{SR}}}/T} \right]$



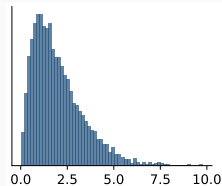
All of this rests on the IID–Normal assumption.

Real returns are not IID Normal — Cont's stylized facts

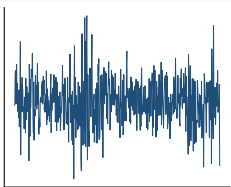
Heavy tails



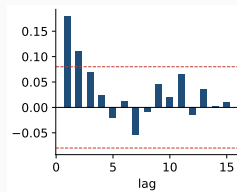
Skewness



Volatility clustering



Autocorrelation (low freq.)



Each breaks an assumption behind $1 + \frac{1}{2}SR^2$.

Prior work: one variance formula per model

Model	$V_{\widehat{SR}}$
IID Normal ^a	$1 + \frac{1}{2}SR^2$
IID Non-Normal ^b	$1 - \gamma SR + \frac{\kappa-1}{4}SR^2$
AR(1) ^c	$\frac{1+\phi}{1-\phi} - \frac{1+\phi+\phi^2}{1-\phi^2}\gamma SR + \frac{1+\phi^2}{1-\phi^2}\frac{\kappa-1}{4}SR^2$
GARCH(1,1) ^d	$1 - \frac{1-\beta}{1-\alpha-\beta}\gamma SR + \frac{\kappa-1}{4}\frac{(1-\beta)^2(1+\alpha+\beta)}{(1-\alpha-\beta)(1-2\alpha\beta-\beta^2)}SR^2$

The point estimate $\widehat{SR}$ is identical in every row — only the variance (interval width) changes.

γ = skewness, κ = kurtosis.

^a (Lo 2002) ^b (Mertens 2002) ^c (López de Prado, Lipton, and Zoonekynd 2025) ^d (López de Prado, Porcu, et al. 2026)

Contribution: closing the gap — AR(1)-GARCH(1,1)

No closed form existed when **both** features are present: AR in the mean **and** GARCH in the variance.

Asymptotic variance, AR(1)-GARCH(1,1), general innovations

$$V_{\widehat{SR}} = \frac{1 + \phi}{1 - \phi} - b\gamma SR + \frac{1}{4}c SR^2$$

Here b and c pack every serial-dependence and heteroskedasticity effect into the interval width; their full expressions are on the next slide.

Method of moments + delta method; all dependence packed into the long-run variance matrix S .

Semi-closed-form: odd moments of σ_t^2 via Taylor expansion.

Lo, Mertens, AR(1), GARCH(1,1) all recovered as restrictions ($\phi = 0$ or $\alpha = \beta = 0$).

AR(1)-GARCH(1,1): the coefficients in full

Skewness channel (the $b\gamma$ SR term), through S_{12} :

$$S_{12} = \frac{\sigma^3 \gamma (1 + \phi + \phi^2)}{(1 - \phi^2)(1 + 2\phi\alpha - \phi\beta)} \left[2\phi\alpha + \frac{(1 - \beta)(1 - \phi(\alpha + \beta))}{1 - \alpha - \beta} \right]$$

Kurtosis coefficient: $c = \frac{1}{1 - \phi^2} \left(4\phi^2 + \kappa \frac{\Sigma}{1 + X} + \gamma^2 Z \left(N - \frac{M}{1 + X} \Sigma \right) \right)$

Auxiliary terms:

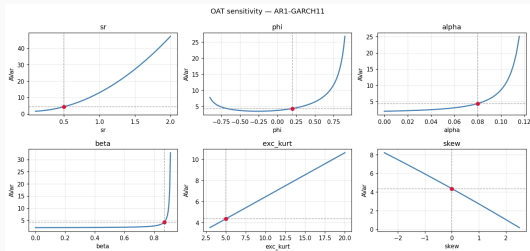
$$M = 1 + \phi^2(2\alpha - \beta), \quad N = 2\phi^2\alpha + \frac{1 - \beta}{1 - \alpha - \beta} (1 - \phi^2(\alpha + \beta))$$
$$\Sigma = \frac{2}{3}X + \frac{(1 - \beta)^2}{(1 - \alpha - \beta)^2} \frac{1 - (\alpha + \beta)^2}{1 - 2\alpha\beta - \beta^2}, \quad Z = \frac{2}{3}\alpha \frac{(1 - \phi(\alpha + \beta))(1 + \phi + \phi^2)^2}{(1 + \phi)^2(1 + 2\phi\alpha - \phi\beta)^2} \frac{4\phi}{1 - \phi^2(\alpha + \beta)}$$

Explicit asymptotic variance (auxiliary terms substituted)

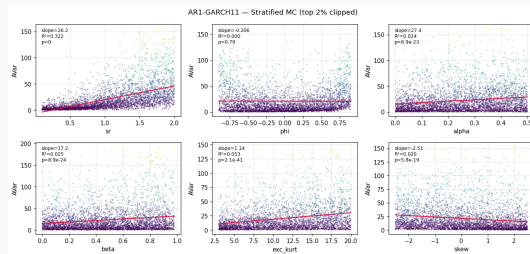
$$V_{\widehat{\text{SR}}} = \frac{1 + \phi}{1 - \phi} - b\gamma \text{SR} + \frac{\text{SR}^2}{4(1 - \phi^2)} \left(4\phi^2 + \kappa \frac{\Sigma}{1 + X} + \gamma^2 Z \left(N - \frac{M}{1 + X} \Sigma \right) \right)$$

γ = skewness, κ = kurtosis, σ , X as defined in the text.

Which parameters move the variance?



One-at-a-time sweep



Axis-stratified Monte Carlo

- **SR is the dominant driver** (convex, quadratic).
- Kurtosis & skewness next (steady, $\sim$ linear).
- ϕ, α, β : mild on average, **explosive near the boundaries** ($\phi \rightarrow 1, \alpha + \beta \rightarrow 1$).

**Do the confidence intervals
actually work?**

Testing the intervals: coverage on synthetic data

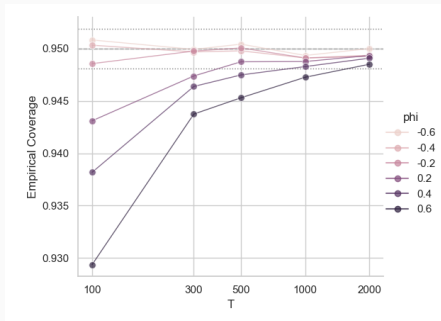
Simulate from a known DGP; measure how often the 95% interval contains the true SR.

Why coverage, not variance?

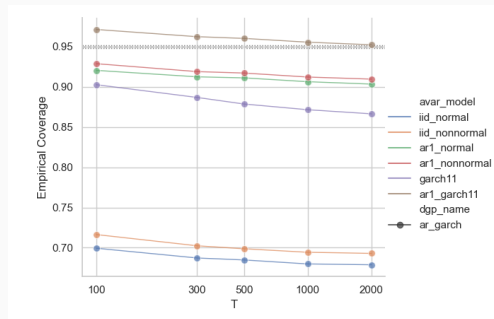
- It is the **inferential target** the user cares about.
- It is a **joint test** — catches bad variance, bias, *and* non-normality.
- It has a **calibrated target**: a binomial acceptance band around 0.95.

Curve below the band $\Rightarrow$ intervals too narrow $\Rightarrow$ overconfident.

Two findings: sign asymmetry & the AR-GARCH gap

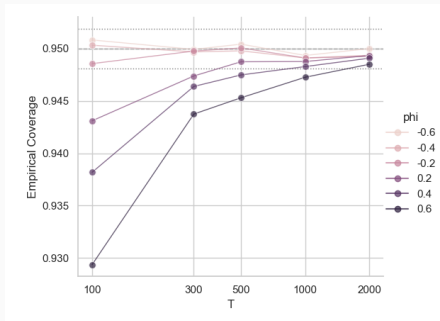


AR(1) DGP: $\rho = +0.6$ vs $\rho = -0.6$

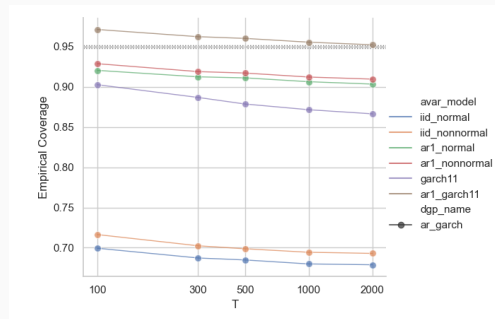


AR-GARCH DGP: all candidate models

Two findings: sign asymmetry & the AR-GARCH gap



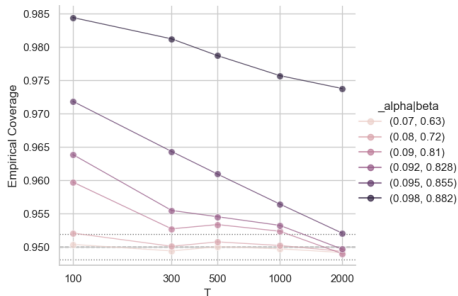
AR(1) DGP: $\rho = +0.6$ vs $\rho = -0.6$



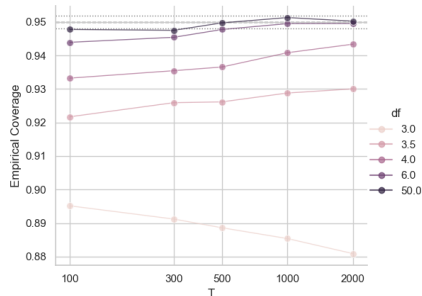
AR-GARCH DGP: all candidate models

- Autocorrelation is **asymmetric in sign**: positive $\rho \Rightarrow$ coverage ≈ 0.69 ; negative $\rho \Rightarrow \approx 0.99$.
- With both stylized facts present, **only the AR-GARCH** formula closes the gap.

Where the asymptotics strain



GARCH persistence $\alpha + \beta$



Student-t degrees of freedom df

- Higher GARCH persistence $\Rightarrow$ **slower** convergence to nominal.
- **No fourth moment** ($df < 4$): coverage **drifts away** as T grows.
- Matches Kao et al. (2025): rate T^φ with $\varphi = 1 - 2/\kappa < \frac{1}{2}$ — $\sqrt{T}$ breaks down.

Is the point estimate even right?

The point estimate is biased — and a popular fix is wrong

$\widehat{SR} = \hat{\mu}/\hat{\sigma}$ is biased in finite samples. Christie (2005) assumes $\hat{\sigma}$ unbiased, but by Jensen (concavity of $\sqrt{\cdot}$):

$$\mathbb{E}[\hat{\sigma}] = \mathbb{E}[\sqrt{\hat{\sigma}^2}] < \sqrt{\mathbb{E}[\hat{\sigma}^2]} \leq \sigma$$

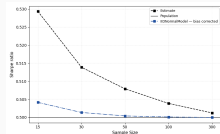
Christie: $SR(1 + \frac{1}{2T})$ — **too small**. Correct: $SR(1 + \frac{3}{4T})$. General correction via the same S :

$$\mathbb{E}[\widehat{SR}] - SR \approx \frac{SR}{2(T-1)} \left(\frac{S_{11}}{\sigma^2} - 1 \right) + \frac{1}{2T} \left(-\frac{S_{12}}{\sigma^3} + \frac{3SR}{4\sigma^4} S_{22} \right)$$

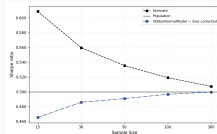
Model	Corrected estimator $\widehat{SR}_{\text{corr}}$
IID Normal	$\widehat{SR} \left(1 + \frac{3}{4T}\right)^{-1}$
GARCH(1,1)	$\left(\widehat{SR} + \frac{\gamma}{2T} \frac{1-\beta}{1-\alpha-\beta}\right) \left(1 + \frac{3(\kappa-1)}{8T} \frac{(1-\beta)^2(1+\alpha+\beta)}{(1-\alpha-\beta)(1-2\alpha\beta-\beta^2)}\right)^{-1}$
AR(1)-GARCH(1,1)	$\left(\widehat{SR} + \frac{1}{2T} \frac{S_{12}}{\sigma^3}\right) \left[1 + \frac{1}{2(T-1)} \left(\frac{S_{11}}{\sigma^2} - 1\right) + \frac{3}{8T\sigma^4} S_{22}\right]^{-1}$

First analytical bias corrections under GARCH and AR-GARCH.

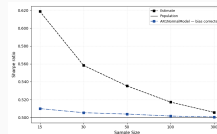
The corrections converge faster



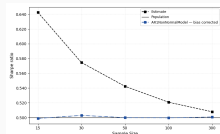
(a) IID Normal



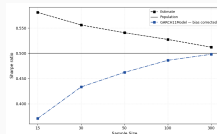
(b) IID Non-Normal



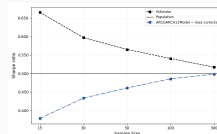
(c) AR(1) Normal



(d) AR(1) Non-Normal



(e) GARCH(1,1)



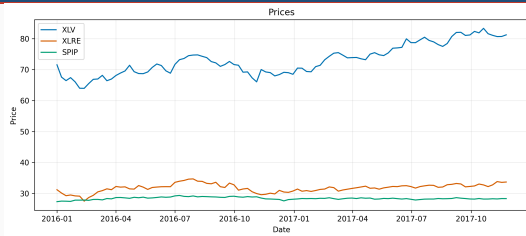
(f) AR(1)-GARCH(1,1)

Figure 1: Individual comparison of the improvement from the bias correction.

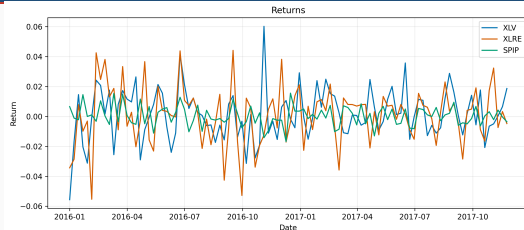
Corrected estimator (blue) converges faster in every model. GARCH is slightly off at very small T , then overtakes — reported honestly.

Does the framework survive real data?

Real data: three SPDR ETFs (2016–2025, weekly)



Price levels



Weekly log-returns

Chosen because they show **both** serial correlation in the level and conditional heteroskedasticity.

Series	Ljung–Box (level)	ARCH-LM	Normality (JB)	Selected specification
XLV	reject **	reject ***	reject ***	AR(1)-GARCH(1,1)-skew- t
XLRE	reject ***	reject ***	reject ***	AR(1)-GARCH(1,1)-skew- t
SPIP	reject ***	reject ***	reject ***	AR(1)-GARCH(1,1)-skew- t

Hierarchical testing (Ljung–Box $\rightarrow$ Engle LM $\rightarrow$ JB/symmetry) selects the same model for all three.

Inference across models: the Probabilistic Sharpe Ratio

$$\widehat{\text{PSR}}(\text{SR}^*) = \Phi\left(\frac{\widehat{\text{SR}} - \text{SR}^*}{\sqrt{V_{\mathcal{M}}(\hat{\theta})/T}}\right), \quad \text{SR}^* = 0$$

Same $\widehat{\text{SR}}$ in every column; only $V_{\mathcal{M}}$ changes.

Series	iid Normal	AR(1)	GARCH(1,1)	AR(1)-GARCH(1,1)
XLV	95.80	96.95	94.94	96.22
XLRE	76.51	79.89	76.14	78.72
SPIP	42.88	42.20	42.80	42.21

XLV crosses the 95% line; the correction is **non-monotone** in model complexity.

Comparing PSRs across columns is not a formal model test — each uses a different reference distribution. It mirrors an analyst committing to one specification.

Beyond a single number: pairing with VaR

Beyond one metric: joint distribution with Value-at-Risk

Non-parametric (empirical quantile)

$$V_h^{np} = \begin{pmatrix} 1 + \frac{SR^2}{2} & \frac{\mu z_\alpha}{2} - \sigma \\ \frac{\mu z_\alpha}{2} - \sigma & \sigma^2 \frac{\alpha(1-\alpha)}{\varphi(z_\alpha)^2} \end{pmatrix}$$

- **Negative covariance** ($\frac{\mu z_\alpha}{2} - \sigma < 0$): a shock that raises $\widehat{SR}$ also lowers $\widehat{VaR}$ — they err together.
- **Only the (2, 2) entry differs.** Parametric is more efficient; the gap grows in the tail.

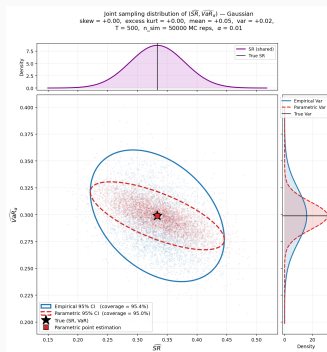
$\varphi(\cdot)$ = standard normal density, $z_\alpha = \Phi^{-1}(\alpha)$.

Parametric (Gaussian)

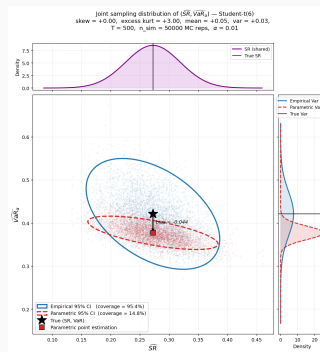
$$V_h^p = \begin{pmatrix} 1 + \frac{SR^2}{2} & \frac{\mu z_\alpha}{2} - \sigma \\ \frac{\mu z_\alpha}{2} - \sigma & \sigma^2 \left(1 + \frac{z_\alpha^2}{2}\right) \end{pmatrix}$$

α	0.05	0.01	0.001
V_{22}^{np} / V_{22}^p	≈ 1.90	≈ 3.77	≈ 11

Efficiency vs robustness, in one picture



Gaussian returns: parametric ellipse **inside** the non-parametric one. Coverage 95.0% vs 95.4%.



Student- t_6 returns: parametric ellipse **biased** (shift -0.044), coverage **14.8%**; non-parametric stays **95.4%**.

Parametric wins when the distribution is right; fails badly when it is wrong.

How reliable is this number?

1. **Precision** depends on the model assumed for returns.
2. The **wrong model** makes intervals lie — usually too narrow.
3. In **small samples** the estimate is biased — the popular fix is too small.
4. Even when **correctly specified**, parameters near the **boundaries** (ϕ , $\alpha + \beta$ persistence) cause trouble.

A single framework — built on the long-run variance matrix S — answers all four for the AR-GARCH world real returns live in.

Thank you — questions welcome.

Code: github.com/Fabro2701/sharpe_ratio_TFM